

Ramoudane Orou Sannou, PhD

SUSTAINABILITY SCIENCE, DATA SYSTEMS DESIGN, RISK ANALYTICS & APPLIED RESEARCH

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Applied researcher and agricultural socioeconomist with over eight years of experience designing and owning agricultural and environmental data systems in low- and middle-income countries. At IFC (World Bank Group), owns the global Contextual Risk Index end-to-end, including indicator taxonomy, source hierarchy, validation rules and automated R workflows for environmental, social and climate-fragility screening across 200+ economies. Independently owns the conceptual design of IntegralDD, a multi-module risk-data platform integrating a dozen-plus external sources under real licensing and data-governance constraints. Built a 12-domain, 47-indicator analytical framework and data-quality protocol for UNFPA across GCC countries and designed a national survey instruments behind civil registration and vital statistics systems in Burkina Faso. PhD summa cum laude (TU Dresden, Germany) focusing on sustainability indicators for agri-food systems; MSc & BSc in Agronomy. Comfortable translating field constraints into schemas engineers can build against.

RESEARCH & PROFESSIONAL EXPERIENCE

Contextual & E&S Risks — International Finance Corporation (World Bank Group)

02/2024 – present

E&S Policy & Risk Department and FCS Africa Team

- Lead the development, maintenance and continuous improvement of IFC's global Contextual Risk Index, owning the end-to-end data and analytical workflow, including sourcing, integrating, cleaning and reconciling data from multiple internal and external databases, and building automated data-processing workflows in R.
- Apply the index methodology end-to-end (data normalization and standardization, indicator weighting, aggregation and index-score calculation) with validation and consistency checks to ensure the robustness, reproducibility and integrity of the resulting risk scores before shipping on Power BI.
- Developed conflict-sensitivity and E&S risk-screening frameworks for African frontier-market investments in agribusiness, solar and manufacturing, integrating geospatial data from external sources (ACLED, UNHCR) to map fragility-risk concentrations and inform investment-viability thresholds.
- Led the technical response to an external audit of the Contextual Risk Index by EY, and now leading its redesign and full automation with a third-party technology firm, including contributing to the firm's selection and overseeing technical requirements for the new solution.

Research Fellow — United Nations University, UNU Resource Nexus

11/2025 – present

Doctoral Researcher 05/2021 – 04/2025

- Conducted a systematic literature review of 128 scientific studies to build a structured taxonomy of social sustainability indicators for agri-food systems, as a centralised framework organising previously dispersed social-sustainability metrics into a coherent indicator set (Sustainable Production and Consumption, 2023).
- Researched what sustains reliable farmer participation in cooperative innovations — and what erodes it — through a mixed-methods case study of shea-nut cooperative members in northern Ghana, identifying the organisational and cultural conditions under which smallholders adopt and sustain participation in new arrangements (International Journal of Agricultural Sustainability, 2025).
- Built the analysis-ready dataset behind a study of forest-based land restoration and food security across the African countries, harmonising FAO survey microdata, deep-learning land-cover classifications derived from Sentinel-2 imagery in Google Earth Engine, and national socio-economic panel indicators into one geographically and temporally aligned panel (Land Use Policy, 2025).

IntegralDD — Independent Data Platform (in development)

06/2026 – present

Self-directed build of a due-diligence data platform for institutional investors evaluating frontier-market risk; Working MVP

- Own the end-to-end conceptual design of a multi-module subnational risk-scoring platform (conflict, climate, governance and natural-resource risk) and built it through to a working MVP covering national-scale pilots in Benin (77 communes) and Tunisia (264 délégations), including dataset prioritization, module/schema design and interoperability across a dozen-plus external sources (ACLED, WRI Aqueduct, NASA POWER, IOM DTM, GDELT, ReliefWeb, VIIRS/EOG).
- Led a systematic data-quality and calibration audit across four scoring modules, finding and fixing real defects, including data-integrity bugs.
- Navigate data governance and licensing constraints directly, commercial-use terms for conflict-event data, manual API approval processes, client-credentialing requirements, as a standing part of onboarding each new data source.

Freelance Consultant — Numbers and People Synergy (NAPS)

2019 – present

Selected freelance assignments, remote

- Statistics and data lead for a NAPS assignment to UNFPA's Gulf Cooperation Council Office: designed a 12-domain, 47-indicator analytical framework and data-quality protocol for Country Profiles on Persons with Disabilities across six GCC countries, operationalised into a live data-collection and dashboard system (*in progress*).
- Developed and applied imputation algorithms for cross-country comparability of indicators across several regional indices in Africa, Asia and Australia.

Data Fellow, International Specialist — United Nations Population Fund

09/2019 – 04/2021

- Designed national survey instruments and statistical models for civil registration and vital statistics systems, and delivered the analytical training that secured government adoption in Burkina Faso.
- Managed a private-sector technology rollout for digital birth registration, reaching 3,000+ registrations across three pilot villages in six months, coordinating collection protocols and record quality with government counterparts.

Research Analyst, Independent Office of Evaluation — International Fund for Agricultural Development (IFAD)

04/2021 – 09/2021

- Contributed to a sub-regional evaluation of IFAD agricultural investment programmes across six fragile countries in the G5 Sahel and Northern Nigeria, collecting data, analyzing and reporting.

Researcher — African Union Commission

08/2018 – 08/2019

- Collected, harmonised and analysed data from national statistical offices across all 55 African Union member states, consolidating inputs into a unified continental monitoring framework for the State of African Youth Report 2019.

EDUCATION**PhD, Integrated Management of Environmental Resources — Technical University of Dresden, Germany**

2021 – 2025

Thesis: The Social Dimension of Sustainability in Agri-food Systems (summa cum laude) · Chair of Environmental Management and Accounting, with United Nations University (UNU Resource Nexus)

Postgraduate Diploma, Environmental Management — Technical University of Dresden (Best Final Paper Award)

2018

MSc & BSc, Agronomy — Rural Economics & Sociology — University of Parakou, Benin

2011 – 2018

PEER-REVIEWED PUBLICATIONS (Selected)

Orou Sannou, R., Schneider, C. (2026). Security shaping sustainability: the difference between securitised and community-based conservation in Benin. *Sustainable Development*.

Orou Sannou, R., Günther, E. (2025). Exploring the resource nexus between forest-based land restoration and food security: the case of the African Great Green Wall initiative countries. *Land Use Policy*, 151, 107499.

Orou Sannou, R., Schneider, C., Walz, M., Günther, E. (2025). Organisational innovations and access to sustainable finance in agriculture: the case of shea farmers in Ghana. *International Journal of Agricultural Sustainability*, 23(1).

Guzman, J., **Orou Sannou, R.,** Caucci, S. (2025). Growing sustainability: socio-ecological drivers of urban and peri-urban agriculture under multiple framework assessments. A review. *Journal of Urban Management*.

Orou Sannou, R., Kirschke, S., Günther, E. (2023). Integrating the social perspective into the sustainability assessment of agri-food systems: a review of indicators. *Sustainable Production and Consumption*, 39, 175–190.

PRACTITIONER & POLICY OUTPUTS

Orou Sannou, R., Raja, A., Hammami, H., Ranjan, R., Serano, C. A. (2025). Empowering women through agroforestry: breaking financial barriers for community resilience. UNU Resource Nexus policy brief.

Orou Sannou, R. (2025). Shifting climate diplomacy: positive narratives for accelerated action. UNU-CRIS.

METHODS & SUBSTANTIVE EXPERTISE

Data systems & governance — end-to-end platform and schema design · indicator taxonomy and metadata standards · multi-source ingestion and interoperability (survey, geospatial, administrative, event data) · data-quality audit, validation and calibration protocols · provenance and source onboarding · data licensing, access and ethical clearance

Quantitative & environmental data — panel and archival econometrics in R · remote sensing and deep-learning land-cover classification (Sentinel-2, Google Earth Engine) · digital survey design

Tools — R (advanced) · Python · SQL · MaxQDA · ArcGIS · Google Earth Engine · Power BI

TEACHING, DISSEMINATION & SERVICE

- Workshop lead, Commonwealth Data Science & Innovation Workshops, Africa and the Caribbean (2022, 2023).
- Subject Matter Expert, Global Reporting Initiative — peer-reviewed the French translation of GRI 13, the international sector standard for agriculture, aquaculture and fishing (2023).

HONOURS, LANGUAGES & CERTIFICATIONS

Honours — Best Final Paper Award, CIPSEM, TU Dresden (2018) · Government of Benin Excellence Scholarship (2011)

Languages — French (native) · English (fluent, TOEFL 107) · German (B1, Goethe-Zertifikat) · Arabic (basic) | **Certifications** — Yale University (Financing & Deploying Clean Energy) · GARP (Sustainability & Climate Risk - SCR) · Frankfurt School (Certified Expert in Sustainable Finance)